

EBU6505 – Reasoning and Agents

Block 2, Day 4 – Planning with Uncertainty

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Teaching Week 4

What have we covered this week?

- Day 1: Bayesian networks for static uncertain reasoning
- Day 2: exact inference with enumeration and variable elimination
- Day 3: temporal models such as hidden Markov models (HMMs), Kalman filters, and dynamic Bayesian networks (DBNs)

Main idea: So far we have focused on belief. Today we move from belief to action.

What are we going to cover this week?

- Day 1: Bayesian networks
- Day 2: exact inference in Bayesian networks
- Day 3: probabilistic reasoning over time
- Day 4: planning with uncertainty
- Day 5: dynamic decision network (DDN) and partially observable Markov decision process (POMDP) synthesis

Main idea: This week moves from uncertain belief, to temporal belief, and then to decision-making.

What are we going to cover today?

- utility and expected utility
- one-off decisions and decision networks
- sequential decisions with Markov decision process (MDP) planning
- value iteration and policy iteration
- why partial observability leads to belief-state planning

Main idea: The lecture has one simple flow: preferences, decisions, planning, then the bridge to partially observable Markov decision processes.

- *Artificial Intelligence: A Modern Approach*, 4th ed.
Chapter 16: Making Simple Decisions
Chapter 17: Making Complex Decisions
- *Artificial Intelligence: Foundations of Computational Agents*, 3rd ed.
Chapter 12: Planning with Uncertainty

Main idea: These readings match today's flow: one-off decisions first, then sequential decision processes.

Learning Outcomes for Today

By the end of this session, you should be able to:

- explain utility and expected utility in simple language
- distinguish one-off decisions from sequential decisions
- explain planning in a Markov decision process (MDP) with a known model
- compare value iteration and policy iteration
- explain why partial observability leads to belief-state planning

Why Plan Under Uncertainty?

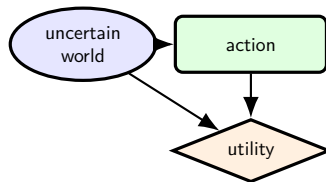
Question: Why is choosing an action hard when the world is uncertain?

Reality

the true world state may be uncertain
actions may have uncertain results

Consequence

a good action now can create better future choices later



Main idea: Planning under uncertainty means choosing actions that are good on average.

What Is Utility?

Question: How do we tell the model which outcomes we prefer?

Simple meaning

Utility is a score for how good or bad an outcome is.

Example

A robot may prefer fast delivery, low energy cost, and safe motion.

outcome	utility
fast and safe delivery	10
safe but slow delivery	7
delay or extra cost	3
unsafe outcome	-20

Main idea: Utility turns vague preferences into numbers the planner can compare.

Expected Utility

Question: How do we compare actions when each action can lead to more than one outcome?

$$EU(a) = \sum_{s'} P(s' \mid a) U(s')$$

Symbols used on this slide

$EU(a)$: expected utility of action a

$P(s' \mid a)$: probability of possible outcome s'

$U(s')$: utility of that outcome

Interpretation

Multiply each possible outcome by how likely it is, then add them together.

Main idea: Expected utility is the average score of an action after uncertainty is taken into account.

A Small Expected-Utility Example

Question: Which route should the delivery robot choose?

action	possible outcome	probability	utility
fast route	clear road	0.6	10
fast route	traffic delay	0.4	2
safe route	normal arrival	1.0	7

Fast route

$$EU = 0.6(10) + 0.4(2) = 6.8$$

Safe route

$$EU = 1.0(7) = 7$$

Main idea: The safe route wins because its average utility is slightly higher.

What Is the Value of Information?

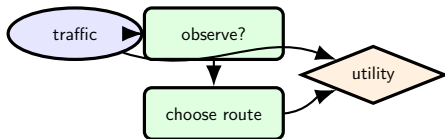
Question: When is it worth spending time or cost to learn more before acting?

Without extra information

choose one action now using current uncertainty

With extra information

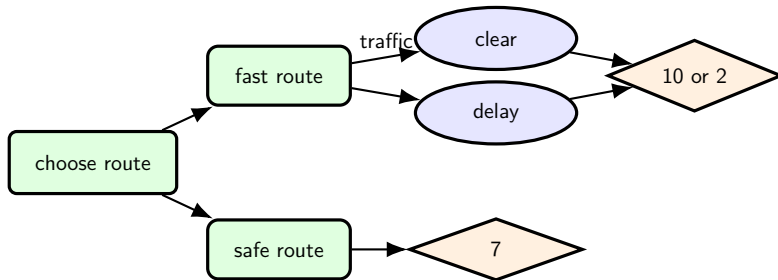
observe first, reduce uncertainty, then choose an action



Main idea: Information has value when better knowledge can change the action and improve utility.

Decision Trees

Question: How can we write down a simple one-off decision problem step by step?



Main idea: A decision tree makes the order of decision, uncertainty, and utility explicit.

Decision Networks

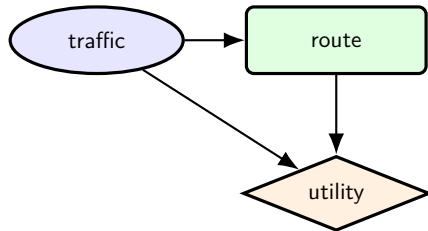
Question: How do we show the same problem more compactly?

Node meanings

Chance node: uncertain part of the world

Decision node: choice the agent can make

Utility node: final score of the result



Main idea: A decision network shows the same logic as a tree in a compact graph.

One-Off Decisions vs Sequential Decisions

Question: What changes when the agent acts more than once?

One-off decision

- one action
- one uncertainty calculation
- choose once

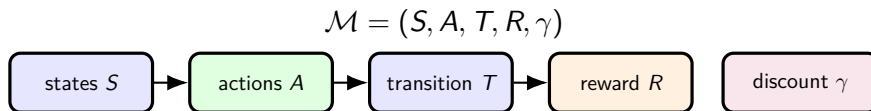
Sequential decision

- many actions over time
- current action changes future state
- need a policy

Main idea: Sequential planning is harder because the agent must care about future consequences, not only the next step.

MDP Components Refresher

Question: Which five pieces define a planning problem as a Markov decision process (MDP)?



Symbols used on this slide

S : possible states

A : possible actions

$T(s, a, s')$: probability of next state

$R(s, a, s')$: immediate reward

γ : how much future reward matters

Main idea: A Markov decision process (MDP) is the standard model for sequential planning when the state is fully observable.

Planning in an MDP Is Not RL

Question: What is the difference between planning in a Markov decision process and learning in one?

Planning

the model is known
compute good values and policies from
the model

Reinforcement learning

the model is unknown
learn from experience and feedback

Main idea: Today we are not learning the model. We are planning with a model that is already given.

What Is the Planner Maximising?

Question: What is the long-term quantity the planner wants to make as large as possible?

$$G_t = \sum_{k=0}^{\infty} \gamma^k r_{t+k+1}$$

Symbols used on this slide

G_t : total reward from time t onward

r_{t+k+1} : later reward

γ : weight on future reward

Interpretation

high γ : future reward matters more

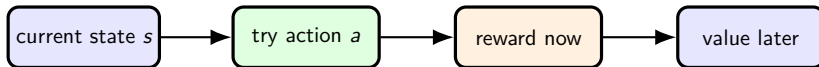
low γ : the planner is more short-term

Main idea: The planner aims to maximise expected long-term return, not just the next reward.

Bellman Backup Intuition

Question: How does the planner compare one action against another in one state?

$$V^*(s) = \max_a \sum_{s'} T(s, a, s') [R(s, a, s') + \gamma V^*(s')]$$



Symbols used on this slide

$V^*(s)$: best long-term value of state s

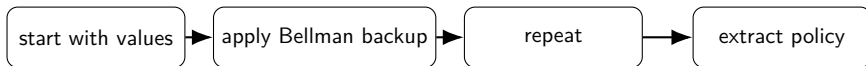
$T(s, a, s')$: chance of each next state

$R(s, a, s')$: immediate reward for that transition

Main idea: Bellman backup asks: which action gives the best mix of reward now and value later?

Value Iteration

Question: How does value iteration solve the planning problem?



What changes?

values for each state

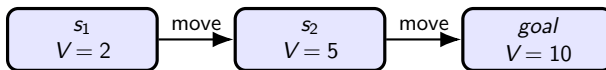
When do we stop?

when the values change only a little

Main idea: Value iteration improves state values first, then reads the best action from those values.

A Tiny Value-Iteration Picture

Question: What does repeated improvement look like in a very small state space?

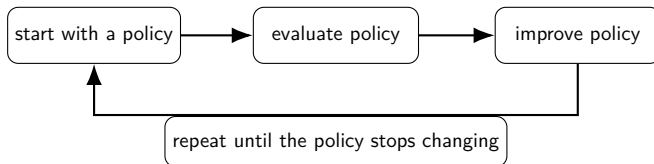


values rise as good future states push value backward

Main idea: Good future states send value backward to earlier states through repeated backups.

Policy Iteration

Question: How does policy iteration differ from value iteration?



Simple language

First ask, "How good is this current policy?" Then ask, "Can I improve any action?"

Main idea: Policy iteration alternates between judging one policy and improving it.

Value Iteration vs Policy Iteration

Question: How can we compare the two planning algorithms quickly?

feature	value iteration	policy iteration
main update	update values directly	evaluate then improve a policy
starting point	any initial values	any initial policy
common intuition	push values backward	polish a policy step by step

Main idea: Both algorithms solve the same planning problem, but they organise the work differently.

What Assumption Does an MDP Make?

Question: What does the planner assume it knows at decision time?

MDP assumption

the agent knows the true current state

Problem

in real systems the observation may be noisy, delayed, or incomplete

Main idea: MDP planning works well only when the current state is fully observable.

When Full Observability Breaks

Question: What does the agent use if it cannot see the true state directly?



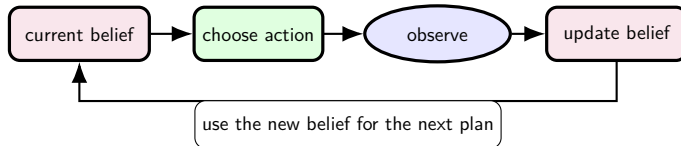
Simple meaning

The agent stores a probability distribution over states instead of one certain state label.

Main idea: Under partial observability, the planner acts on beliefs about state, not on state itself.

Belief-State Planning Loop

Question: How does the agent keep planning when it only has a belief?



Main idea: The loop is: plan, act, observe, update belief, then plan again.

Question: What does a partially observable Markov decision process (POMDP) add on top of an MDP?

MDP

states, actions, transitions, rewards, discount

POMDP

all of the above, plus hidden state and an observation model

Symbols used on this slide

$b_t(s) = P(s_t = s \mid o_{1:t}, a_{1:t-1})$: belief that state s is true at time t

Main idea: A partially observable Markov decision process (POMDP) is a planning model for acting under uncertainty when the true state is not directly visible.

Which Model Fits Which Situation?

Question: How do we know whether we need a decision network, an MDP, or a POMDP?

model	best for	simple description
decision network	one-off decision	choose once under uncertainty
MDP	sequential, fully observed	choose many times with known current state
POMDP	sequential, partially observed	choose many times using belief instead of true state

Main idea: The right model depends on whether decisions are one-off or sequential, and whether the state is visible or hidden.

A Simple Running Scenario

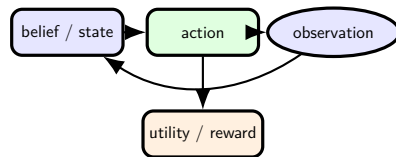
Question: How do all these models appear in one delivery-robot story?

Three views of the same story

Decision network: choose a route once under uncertain traffic

MDP: choose move, wait, or recharge over many steps

POMDP: plan with belief when battery or traffic is only partly visible



Main idea: These are different views of the same uncertain decision problem.

What Can Go Wrong?

Question: Why can planning under uncertainty still fail?

Common problems

- wrong model
- wrong utilities
- missing observations
- overconfidence in noisy evidence

Why it matters

the planner may choose an action that looks optimal in the model but is poor in the real world

Main idea: A mathematically correct planner can still make bad decisions if the model or utility is wrong.

Simple Safety Checks

Question: How do we make uncertain planning safer in real systems?

Model-side checks

- calibrate probabilities
- monitor model drift
- test utility assumptions

Decision-side checks

- keep a safe fallback action
- ask for more evidence when risk is high
- use human approval for costly actions

Main idea: Good planning under uncertainty needs both a good model and sensible safety controls.

Where Are We Going Next?

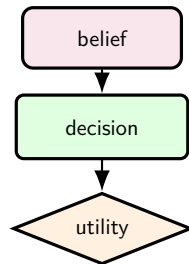
Question: What changes when the uncertainty is both temporal and partially observed?

Today

probabilities, utilities, decision networks,
and MDP planning

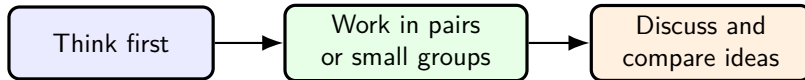
Next lecture

integrate temporal belief, decision nodes,
and POMDP ideas into one synthesis



Main idea: Day 5 combines belief over time with decision-making under uncertainty.

In-Class Exercises



Use this time to work on today's in-class tasks and prepare one clear answer you can discuss.

Main idea: Use the exercise time to decide which planning model fits each scenario and explain your choice clearly.

Summary

- utility and expected utility support one-off decision-making under uncertainty
- decision networks give a compact graphical view of decisions plus uncertainty
- MDPs support sequential planning when the current state is known
- value iteration and policy iteration are two standard planning algorithms
- POMDPs are needed when the true state is hidden and the agent must plan with belief

Main idea: Planning with uncertainty means connecting probability, preference, and future consequences in one decision model.

Further Study

- Read Poole Chapter 12 and AIMA Chapters 16–17.
- Practice: Poole Ch.12, Exercises 12.2, 12.3, 12.6, 12.8, 12.13, and 12.15.
- AIMA decision-theory exercise page: 16.19, 16.20, 16.21, and 16.22.
- AIMA complex-decisions exercise page: 17.1, 17.8, and 17.14.
- Focus on expected utility, decision-network structure, and the difference between one-off choice and Markov decision process planning.